

Introduction

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Course Information

- [Website](#)
- First Day Handout
- Tech Requirements

Pixels

- Graphing Paper
- “Draw a line from $(1,0)$ to $(4,5)$ ”

- The Cartesian Coordinate System
- The Computer Coordinate System
 - Pixels

Shape Representation

Based on the command (function) from the last slide,

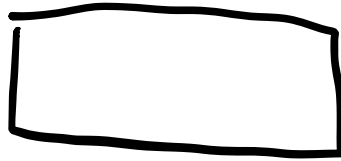
line(1, 0, 4, 5)

think about how you could represent these in English
and in code.

- Rectangle
- Triangle
- Eclipses (Circles)

Points, Lines, Rectangles

- `point(3, 2);`
- `line(1, 2, 7, 3);`
- `rect(1, 2, 5, 4);`



Rectangles

- By default, rectangles are specified by the coordinate for the top left corner of the rectangle, as well as its width and height.
- This can be changed with the function

`rectMode()` ;

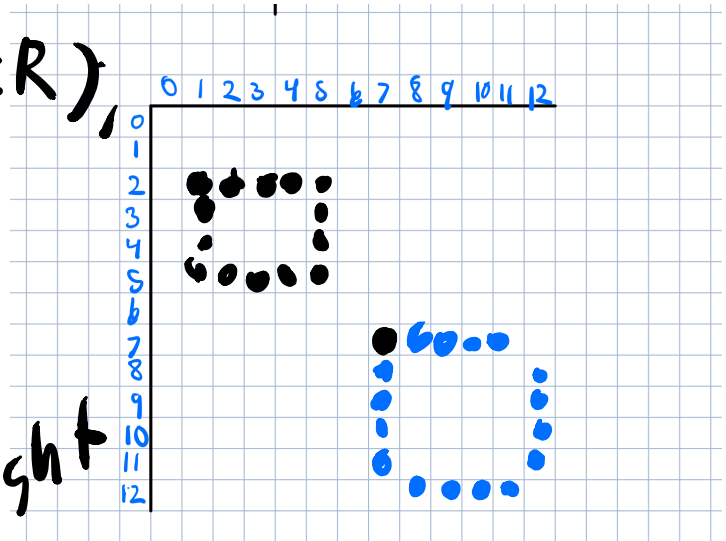
- CORNER, CENTER, CORNER3

`rectMode(CENTER)` ;

`rect(x, y, w, h)` ;

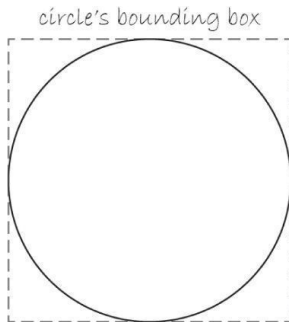
top left

bottom right



Eclipses

- Ellipses are similar to drawing a rectangle, except the default mode is *CENTER*



- Zoomed in, ellipses may seem weird, but it isn't noticable at scale on a display.

Other Shapes

The discussed shapes are only a subset of what processing has to offer. We'll discuss them more in the next chapter.

Color - Grayscale

- A pixel's location is only part of the story. You can also specify it's color.
- Grayscale: Black and White colors defined on a 256 number scale.
- 0 - Black, 255 - White.
- Any other number in between is a shade of gray.
- Colors are represented in the digital world as a byte.

Color - Grayscale

- Computers use binary to represent colors, letters, numbers, sounds, and more.
- Binary has two possible values: 0 and 1
- Each 0 or 1 is a **bit** (**B**inary **D**igit).
- Eight bits make up a byte.
- Think of bits as a switch (0 - Off, 1 - On)
- 0000 0000 = 0 = Black
- 1111 1111 = 255 = White

stroke and fill

- Every shape has a `stroke()` ; and/or `fill()` ;
- Stroke is used to specify the color of an outline or line.
- Fill specifies the inner color of the shape.
- *Lines and point can only use stroke.*
- The background color can also be changed with `background()` ;

stroke and fill

- The default color for `stroke()` ; is black and `fill()` ; is white.
- `stroke`, `fill`, and `background` are used *before* designating the shape.
- Use `noStroke()` ; and `noFill()` ; when no outline or color fill is wanted.

Color - RGB

- Colors are created by combining red, blue, and green.
- RGB color elements also have a range from 0-255.
- Format for specifying color is `fill(red, blue, green) ;`.
 - `fill(255, 0, 0) ;` = Bright Red
 - `fill(0, 255, 0) ;` = Bright Blue
 - `fill(0, 0, 255) ;` = Bright Green

Alpha

- There is an optional fourth component, referred to as “alpha”.
- Alpha means opacity.
- 0 means complete transparency (zero percent opaque).
- 255 is completely opaque (100 percent opaque).

Zoog

